

Time Series Forecasting of Disposable Plastic Waste in Karachi for Circular Economy Application

Mubashra Zulfiqar – Sadia Rehman – Bushra Bakht – Zainab Bibi

Abstract

Rapid urbanization in developing country cities like Karachi (Pakistan) creates plastic waste faster than the waste collection and recycling systems can cope, and short-horizon forecasting is essential for effective planning of circular economies for plastic waste streams, but that is not possible with the existing short to medium term studies, which are mostly aggregate waste or based on the assumption that disruptions are one-off historical events. In order to predict the volume of plastic waste collected in Karachi on a monthly basis for 60 months (January 2020 to December 2024), this study divides the data into training (48 months from January 2020) and holdout (12 months from January 2024) sets. Three forecasting methods are proposed and evaluated in this study: the classical Box-Jenkins ARIMA, an LSTM network, and the decomposable additive-regression model Prophet. Three candidate covariates—per-capita waste generation rate, monsoon season indicator, and indicator of unusual events—were chosen after leakage and redundancy screening. Each model was trained and assessed using the same preprocessing and evaluation techniques using RMSE, MAE, MAPE, and R^2 . The stationarity tests revealed that the target series was stationary, and the ARIMA order was chosen using the out-of-sample information validation criterion. Due to its limited training data, the LSTM performs poorly with a large systematic under-forecasting error ($RMSE = 37.789$ tons, $R^2 = -8.150$), while the ARIMA (0,1,0) with drift model has the lowest holdout error ($RMSE = 3.224$ tons, $MAPE = 0.241\%$, $R^2 = 0.933$), outperforming both the Prophet baseline model and the Prophet with external regressors. The study concludes that the ARIMA model with a simple drift term is adequate to describe the data for this short, gradual, highly trending time series and that there is no benefit to increasing the model's complexity to increase prediction accuracy; however, the prediction intervals may be lower than the risk level in months of disruption during the monsoon or event periods. The study cautions that months of disruption during the monsoon or event periods may have lower-than-expected risk levels in their prediction intervals and recommends that municipal planners use a validated, low-complexity model based on ARIMA's simple drift term for plastic-waste capacity and collection planning.

I. Introduction

Plastic waste is no longer a manageable municipal issue. Recycling, waste governance, and global sustainability policy are currently among the most notable failures of the twenty-first century [32]. Since the start of the twenty-first century, production has roughly doubled to over 430 million tons annually

worldwide [33, 34]. Depending on the methodology employed, the current estimate of the annual generation is between 0 and 400 million tons [33] [34]. Out of this massive quantity, only 9% is recycled [33]. Nearly one-third is burned, dumped in the open, or leaks into rivers due to improper handling at the end of life [33]. This is a highly localized structural imbalance. Only 12 countries produce about 60% of the improperly managed plastic waste [35]. Key plastics-producing nations' opposition to production limits caused negotiations for a legally binding global plastics treaty to stall in August 2025 [32]. This implies that municipal and provincial governments are solely responsible for handling this crisis, with the least prepared cities suffering the most [32]. The consequences of this systemic failure extend beyond landfill boundaries [36]. Decomposing plastic also contributes to methane emissions in landfills, exacerbating the environmental impact of waste management. At street level, the threat is both real and obvious. Plastic bags and packaging clog stormwater systems, causing frequent waterlogging in cities like Karachi. During monsoon floods in August 2025, plastic waste, plastic bottles, and industrial effluent were washed into the Lyari and Malir rivers, as well as directly into the Arabian Sea, increasing the litter footprint of the coastline from 1.5 km to over 2 km offshore [37]. Standing water from these clogged drains can serve as a haven for disease-carrying organisms. When Karachi experienced the worst flooding in nearly a century in 2020, dengue cases immediately increased [38]. Similar flooding in 2022 led to a confirmed cholera outbreak with over 200 cases [39]. The sanitation system's inability to handle the amount of solid waste was the primary cause in both situations [38, 39]. Drainage blockage, urban flooding, and vector-borne diseases are all part of a single causal chain, with plastic waste as the first risk [40].

Circular economy frameworks reverse this destructive cycle by prioritizing recovery strategies. The most advanced method is mechanical recycling, which includes sorting, shredding, and remolding. However, it is still severely limited in places like Karachi. Only 6 tons of the estimated 6,844 tons of plastic waste that the city could recycle each day are actually processed, and most of this is done unofficially [41]. For complex, heterogeneous, and multi-layered plastics, chemical recycling or feedstock can be used instead of mechanical lines to break them down into fuel-grade feedstock or reusable monomers. Waste-to-energy and energy recovery via pyrolysis are growing in popularity on a regional scale in other parts of the world. To justify the capital investment, however, the waste volume and composition must be extremely predictable [42]. When recovery mechanisms are unable to handle it, landfilling is the last option. In Karachi, only 75% of the waste collected is not sent to open landfills like Gondpass and Jam Chakro [41]. Mechanical recycling, chemical recycling, energy recovery, and final landfilling are all separate circular economy applications. Precise knowledge of the quantity of plastic waste entering the pipeline is necessary for each. Predictive modeling becomes a necessity for infrastructure planning due to this operational requirement, rather than just an academic exercise.

Pakistan is a close reflection of Indonesia's structural realities: both are large, rapidly growing plastic producers with policy aspirations that always outweigh policy enforcement [1, 43]. The official and academic estimates of the collection rate of the 14,000-16,500 tons of daily municipal solid waste (MSW) in Karachi are 60-75 percent [43, 44]. Every day, thousands of tons of waste are dumped in rivers and on the streets [43, 44]. Recycling facilities, waste-to-energy facilities, and collection vehicles cannot be sized using national averages under such dynamic conditions. Accurate short-horizon forecasting is a crucial precondition for circular planning, and they require very precise and forward-looking estimates of the plastic waste stream rather than of the trash. Due to this technical requirement, deep learning architectures have replaced traditional statistical models in municipal waste forecasting. Traditional methods such as ARIMA make the assumption of linear and stationary relationships between past and future values. If a waste stream is subject to unpredictable shocks, seasonal disruptions and rapid structural changes, as is common in rapidly urbanizing, developing world cities, this is not the case. Long Short-Term Memory (LSTM) networks are used with gated recurrent architectures to get around this. LSTMs can directly learn long-range, nonlinear temporal dependencies from sequence data, in contrast to ARIMA's stationary assumptions. The data itself presents the second difficulty. The thorough socioeconomic data flows that are typical of wealthier global cities are typically not maintained in developing world metropolises. Due to this shortcoming, sequence models optimized for a limited number of carefully chosen external covariates rather than a complete set of features have been the focus of recent modeling efforts. The data limitation of Bandung, which is based on a single continuous proxy, nightlight radiance, as a proxy for underlying urban economic activity, serves as the foundation for this study [4].

In contrast, this study explicitly combines demographic (population), behavioral (per-capita generation rate), seasonal (monsoon), and shock-event covariates to model the local dynamics of Karachi. Covariates from the city's own administrative and meteorological records are more useful than a single remote-sensing proxy for a city as distinct as Karachi in terms of its seasonal and event calendar. Karachi was selected as the study's focus for a number of particular reasons.

First, estimates of the population range from roughly 16 million in the most recent census to over 23 million based on World Bank accounting [45] [41]. This wide statistical dispersion is highly instructive: Karachi is not an exception to the data-poor, highly volatile baseline that characterizes developing-world metropolises. Second, this urgency is made worse by geography. The city is situated right on the Arabian Sea, and its coastline has been identified as one of the highest hotspots for marine litter in Africa and South Asia. This problem is frequently made worse by flushing events caused by the monsoon, which force plastic inland and into the sea [41, 36]. The predictive integrity of current waste forecasting literature, which frequently concentrates on aggregate, unclassified waste streams [6] or treats structural shocks as isolated, non-recurring events [19], may be impacted by simple mathematical multicollinearity

and feature leakage, two common problems in local datasets. Comparing a time-series framework, benchmarking ARIMA, Prophet, and multivariate LSTM architectures, isolating independent behavioral, demographic, seasonal, and operational shock covariates, and methodically maintaining model integrity against mathematical feature leakage constitute the primary contribution of this study.

II. Literature Review

Global municipal waste is expected to exceed 3.8 billion tons per year by 2050 [1], reviving interest in predictive modeling for urban planning. Statistical models are still the standard in this field and predate deep learning. For instance, regression-based scenario analysis was used to forecast global plastic fates through 2050 [2], while a statistical framework reconciled household waste streams in the Czech Republic without a learning component [3]. Direct comparisons with ARIMA, which was able to compete with LSTM and Prophet in predicting plastic waste collection in Bandung, Indonesia [4] and a classical support vector machine outperformed LSTM when training data was limited in Bogotá [5], complicate the notion that deep learning is always superior. Ethiopian data, on the other hand, produced the opposite results, with LSTM ultimately surpassing both ARIMA and Random Forest [6], and a five-category comparison in Saudi Arabia revealed that both ARIMA and ETS performed poorly on liquid waste because of its volatility [7]. These results directly relate to a study based on a synthetic five-year record rather than ten years of municipal history, indicating that statistical baselines are still helpful in scenarios with little data. However, the mechanism this study examines is still unknown, and none of these studies include a covariate for seasonal or event-driven demand.

In solid-waste time-series research, deep learning architectures and LSTM variants in particular have supplanted the statistical baselines mentioned above. BiLSTM outperformed the standard LSTM, GRU, and 1D-CNN when used to predict nearly 20 years of municipal waste growth in Şırnak, Turkey [8], waste-to-electricity potential in Shanghai, China [9], and the quantity of waste in a smart bin in Australia [10]. The base architecture has been directly altered in a few studies: An enhanced GRU was tested on data from Multan, Pakistan, and other places [12], while an enhanced LSTM with a different activation function and optimizer was applied to Malaysian domestic waste [11]. By combining LSTM with SVM to forecast waste from flooding [13] and integrating LSTM into a production-planning system for the plastics industry [14], hybrid designs go one step further. None of these studies isolate a plastic-specific stream and test its response to seasonal variation, which is the issue this study attempts to solve. Instead, the target variable across this cluster is consistently aggregate municipal waste, an industrial demand figure, or a single physical driver.

A smaller body of research incorporates external driver variables directly into forecasting models, rather than treating waste generation as a univariate series. Prior to using a BiLSTM model to forecast Beijing's

municipal waste, socioeconomic drivers were weighed using Grey Relational Analysis [15]. Similarly, by incorporating waste composition, urbanization indices, and population density, tree-based methods such as XGBoost have surpassed Decision Trees and Random Forests [16]. In order to model construction waste in Tanta, Egypt, a fuzzy neural network was fed land-use classes [17]. A more recent contribution applies this reasoning to a driver that is particularly relevant to plastic: a two-step machine-learning framework was utilized to incorporate hospital visitor counts, which were obtained from anonymized GPS location data, into the forecast of industrial plastic waste collection demand at a Japanese hospital. In one validation month, a Fine Tree model achieved 89.86% accuracy, and the monthly mean accuracy increased from 84.44% to 85.06% [18]. This outcome is benchmarked using MAPE and compared to more than thirty linear, tree-based, kernel, and neural-network models. It demonstrates that, in the absence of a recurrent architecture, a single carefully selected exogenous covariate significantly increases plastic-specific forecasting accuracy. All of the drivers analyzed are land-use, demographic, economic, or single-facility mobility data, and none of them operate outside of a single facility. Nevertheless, when considered collectively, this cluster demonstrates that driver-variable integration is a reliable method to improve accuracy over univariate baselines. This study provides Karachi with a climate-seasonal indicator and a city-scale flag for religious or cultural events, which none of them have.

Waste generation is thought to be sensitive to discrete external shocks, according to a related line of research, which offers the closest analogy to a recurrent cultural event like Eid. During Bahrain's subsequent COVID-19 lockdowns, domestic waste was monitored using a Bayesian-optimized BiLSTM, which outperformed ARIMA in all reported metrics [19]. A similar structure was employed in [13] to forecast the amount of waste produced as a result of flooding by combining LSTM and SVM, with the input variables being the flood depth and rainfall duration. Both show that LSTM models are better at capturing sudden non-stationary changes than statistical models, which is why this study has used an event indicator. However, neither approaches the disruption as a recurrent, calendar-predictable feature; instead, lockdowns and floods are modeled as isolated historical occurrences. A religious festival differs structurally from both situations in that it is held every year on a known but variable date. This is not covered in this literature.

This study follows a parallel methodological approach that compares the performance of LSTM with Linear Regression and Random Forest using RMSE, MAE, MAPE and R^2 . Multiple machine learning models were tested on Chinese provincial data using a Deep Neural Network, and the SHAP values were used to explain the waste generation by GDP and population density [20]. For household forecasting in Saudi Arabia and Brunei, an Optuna-tuned ensemble was benchmarked against nine alternatives, such as SARIMA and XGBoost [21]. In Erode, India, a direct comparison of SVM, Random Forest, and Naive Bayes revealed that plastic and organic waste were the two best indicators of total volume [22]. Using the

same comparative methodology, the Saudi five-category study mentioned above consistently demonstrated that Random Forest, XGBoost, and LSTM outperformed the statistical baselines in every category assessed [7]. This well-known comparison culture is evaluated using the same three-model design as this study, but it does not have a plastic-specific target.

Developing-world evidence is highly relevant here. The Bandung study remains the closest structural match, using LSTM to forecast urban plastic waste [4]. With the exception of seasonal terms and plastic disaggregation, two studies conducted in Multan, Pakistan, used GRU networks to estimate aggregate municipal collection and landfill area from vehicle trips [12, 23]. Regional models in other parts of South Asia include an IoT-driven route optimization system in an unidentified Indian city [26], a ward- and season-aware model in Dhaka that disregarded monsoon-specific rainfall [25], and an LSTM-based routing framework in Surat [24]. Additional frameworks are available for Ethiopia [6] and Egypt [17]. Although this regional cluster confirms active development, it shows that no study has integrated indicators of cultural events, population growth, monsoon seasonality, and a plastic-specific target for any Pakistani city.

The broader body of research on circular economies inspires rather than forecasts methodology. XGBoost had an accuracy of 98.5% for solid waste forecasting and LSTM had an accuracy of 94% for biogas prediction, according to a 128-study PRISMA review that identified AI, IoT, and blockchain as important management technologies [1]. However, by pointing to data scarcity and lack of standardization as persistent challenges, it did support the findings of the Saudi study [7]. Systematic reviews of AI in municipal waste (229 studies) [27] and machine learning in circular supply chains (66 studies) [28] both show this structural inconsistency. A narrative review that places AI and ML in the context of plastics in circular-economy roadmaps does not test an empirical model [29], and LSTM has even been applied to forecast the carbon footprint and strength of plastic mortar [30]. Another study uses LSTM in material science to forecast plastic mortar strength [30], and a different narrative review discusses AI plastics roadmaps without conducting empirical testing [29]. Other uses are still outside-generation modeling or non-empirical. Because plastic and glass and metal are visually similar, the IoT-based CNN classification system's overall accuracy for material identification was only 56% for plastic [31]. This functions as a boundary marker rather than a forecasting standard. The use of a carefully validated synthetic dataset in this study is supported by the fact that, when considered collectively, these studies show that the primary constraints of the field are still data scarcity and standardization.

The literature cited above on municipal waste forecasting either focuses on the aggregate waste stream or considers structural shocks as one-off events. Each of these concerns will be addressed in the proposed work. This research is unique in assessing the performance of an LSTM forecasting architecture that

integrates population growth, monsoon seasonality covariates specifically for a plastic-specific waste stream in a data-scarce Pakistani urban context.

III. Methodology

A. Study Workflow

The research took place in a sequential, and partially parallel, approach from data collection to comparison with a model. After receiving the 60-month dataset of the Karachi municipal waste, the data structures were checked, data timing was performed and rigorous preparation was made for data dependability. The statistical characteristics of the target series and its suggested predictors were then investigated in an exploratory analysis stage, which resulted in two concurrent stages: feature engineering to identify the final set of predictors and split the data chronologically into a training set and a holdout set, and stationarity tests to determine the differencing structure required for ARIMA.

Using the same data, three concurrent development paths—ARIMA, LSTM, and Prophet—each with a model-specific preparation appropriate for the model were conducted. When both approaches produced forecasts for the same twelve-month holdout period and a twelve-month forecast out-of-sample of the observed data, the common forecast generation step took place. A single ranking performance evaluation was produced by aggregating the findings across all model configurations in a comparison analysis stage after all model outputs were compared to the same metrics and holdout objectives. The divergence into parallel model-specific development and the subsequent convergence for a shared assessment ensure that the observed variations in forecasting performance are caused by differences in modeling ability rather than variations in data handling techniques between forecasts. Below are specifics on the dataset utilized in this workflow as well as the preprocessing procedures.

B. Dataset

Based on plastic trash data collected from 60 monthly observations between January 2020 and December 2024, this study forecasts the amount of plastic waste collected by the Pakistani municipality of Karachi. The quantity of plastic waste collected each month (measured in tons) is the dependent variable. There are four candidate predictors along with the target: a generation-rate variable per capita per day, a binary monsoon-season indicator, a binary unusual-event indicator, and population. The dataset is structurally complete: there are no missing calendar months across the five-year period and the dataset is split into a forty-eight-month training set and a holdout set of twelve months representing calendar year 2024.

Previous research on plastic waste collection in Bandung, Indonesia [4] used a single satellite-derived regressor (nighttime radiance), which is similar to the analytical methodology used here: comparative forecasting via ARIMA, LSTM, and Prophet. As an independent application of the same modeling approach to a different research location and variable collection, the current dataset instead uses locally

acquired demographic, seasonal, and operational characteristics unique to Karachi's municipal solid-waste system. This dataset underwent the pretreatment steps outlined below before modeling.

C. Data Preprocessing

To ensure the data set's temporal consistency, dependability, and comparability across all three forecasting methodologies, it was first processed in accordance with a predetermined preprocessing sequence. The records' data types were verified for accuracy, the data was transformed into an appropriate datetime index, and the records were arranged chronologically at one-month intervals. Although no duplicate or missing observations were discovered in the series, a conditional interpolation precaution was maintained for methodological completeness. The interquartile-range approach was used to identify statistical outliers, and the monsoon and event indicators were used to corroborate them. This allowed for the preservation of waste generation changes that might represent useful signals rather than noise. Feature engineering was used to remove all fields from the candidate predictor set that displayed some level of near-zero variance, redundancy, or leakage. Following that, the target series and the retained features (a continuous generation-rate regressor plus two binary indicators of disruptions) were prepared for use in each model in accordance with its specifications, which included formatting as additive decomposition for Prophet, normalization between a minimum and a maximum value for LSTM, and differencing and stationarity testing for ARIMA. There was no information leaking from the "holdout" interval because the scaling and sequence generation parameters were only learned from the training partition. This is a combined, model-ready dataset that was corrected prior to model-specific development, ensuring that any discrepancies in predicting performance would really reflect differences in model performance rather than inconsistent data preparation. On this shared basis, three distinct forecasting techniques—ARIMA, LSTM, and Prophet—were created and are explained in the section that follows.

D. Forecasting Models

The produced dataset was subjected to similar training and holdout circumstances for three forecasting approaches: ARIMA, LSTM, and Prophet. This allowed their comparative performance to be attributed to actual variations in modeling capability rather than inconsistent data preparation.

1) ARIMA

For order (p, d, q), the ARIMA model interprets a differenced time series as a linear function of the series' prior values and the model's prior forecast errors:

$$\varphi(B)w_t = c + \theta(B)\varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2)$$

where w_t is the d-times differenced series, B is the backshift operator, and $\varphi(B)$ and $\theta(B)$ are the autoregressive and moving-average polynomials of order p and q, respectively. A grid search over p, q =

$\{0, 1, 2, 3\}$ with $d = 1$ (fixed by stationarity test) and a drift added to the model at all times was used to pick the order. The models were then scored using AIC and BIC, and each model was tested for validation over the following 12 months using RMSE, MAE, and MAPE. Higher-order candidates tended to match the in-sample data quite well but overfitted in the out-of-sample, which also influenced the final specification. This model turns into a random walk with drift in the chosen model ARIMA (0,1,0) with drift.

$$w_t = c + \varepsilon_t$$

It was chosen due to its parsimony, ability to generalize, and directly interpretable drift coefficient (still roughly 3.22 tons/month) calculated using maximum-likelihood on the 48-month training dataset.

2) Long Short-Term Memory (LSTM)

One-step-ahead forecasting, which predicts a scaled objective for the upcoming month using three multiplicative gates and a cell state with the ability to lessen the gradient over the sequence, is modeled by the LSTM as a supervised sequence-learning problem:

$$\begin{aligned} i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), & f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t) \end{aligned}$$

A dense head was used to generate a scalar prediction from the final concealed state. The Adam optimizer [46] was used to optimize the mean squared error loss with a default patience of 60 and a maximum of 1,000 epochs (one season) before training was completed to match the final output on all 36 train sequences. Since the network's trainable parameters (about 11,000) are significantly greater than the number of training sequences (36), the architecture has been carefully tailored for the "small end" of the standard. Since capacity control is the network's main design difficulty, this is done on purpose.

3) Prophet

The series is additively broken down by Prophet into trend, seasonality, and regressor effects:

$$y(t) = g(t) + s(t) + \sum \beta_x x_x(t) + \varepsilon_t$$

Seasonality as a shortened Fourier series and data-driven changepoints with a piecewise-linear trend. A linear-growth specification (no saturating ceiling) with additive seasonality—but not weekly or daily seasonality—was employed since the series was thought to have a clear monotonic trend. Weekly and daily seasonality were ruled out as unnecessary for monthly data with just four complete seasonal cycles. One baseline with no external regressors and one external-regressor with the external regressors added (marginal addition) to the baseline were fitted using the same parameters (95% prediction interval).

Following the definition of the specifications for each of the three forecasting models, the holdout performance of each model is evaluated and compared using the following criteria.

E. Model Evaluation

The accuracy of the forecasts was assessed using four complementary metrics, which were calculated in the same manner for each model output compared to the holdout series of the previous 12 months: y_i is the actual value, $\hat{y}_i$ is the forecast, $\bar{y}$ is the mean value of the holdout series, and n is the number of values in the holdout series.

The average size of the errors in the target's native units (tons) is known as the Mean Absolute Error (MAE):

$$MAE = (1/n) \sum |y_i - \hat{y}_i|$$

Regardless of the size of the series, the Mean Absolute Percentage Error (MAPE), which is a percentage of the actual value presented without any units on the error measure, can be compared across series:

$$MAPE = (100/n) \sum |(y_i - \hat{y}_i) / y_i|$$

When compared to a naïve forecast, which predicts the mean of the data throughout the hold-out period, the coefficient of determination (R^2) measures how accurate the forecasts are.

$$R^2 = 1 - [\sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2]$$

These four metrics were selected because each captures a distinct property of forecast error: MAE communicates typical error magnitude in interpretable, native units; RMSE, when compared against MAE, reveals whether error is evenly distributed or concentrated in a small number of severe misses; MAPE provides a scale-free measure suitable for benchmarking independent of the series' magnitude; and R^2 is the only one of the four that can indicate a forecast performing worse than a trivial constant-mean baseline, a failure mode invisible to the other metrics alone.

This multi-metric approach follows standard practice in the time-series forecasting literature [47] and parallels the evaluation framework of the source study [4]. All four metrics were computed under an identical procedure across the ARIMA, LSTM, and both Prophet. These four metrics were selected because each captures a distinct property of forecast error: MAE communicates typical error magnitude in interpretable, native units; RMSE, when compared against MAE, reveals whether error is evenly distributed or concentrated in a small number of severe misses; MAPE provides a scale-free measure suitable for benchmarking independent of the series' magnitude; and R^2 is the only one of the four that can indicate a forecast performing worse than a trivial constant-mean baseline, a failure mode invisible to the other metrics alone. configurations, ensuring that the resulting cross-model comparison rests on a uniform and comprehensive basis rather than a single, partial accuracy measure.

IV. Results and Discussion

Three forecasting approaches — ARIMA, LSTM, and Prophet — were applied to Karachi’s monthly plastic-waste collection series (60 months, January 2020–December 2024), using a chronological 48-month training / 12-month holdout (2024) split, following the same evaluation protocol used by [4] for the comparable Bandung, Indonesia case.

A. ARIMA Model

An Augmented Dickey-Fuller (ADF) test on the undifferenced target series returned a statistic of -0.2463 ($p = 0.9328$), failing to reject the null hypothesis of a unit root. A KPSS test confirmed this result (statistic = 1.2872 , $p = 0.01$), and both tests agreed that a single first difference ($d = 1$) achieves stationarity (ADF = -10.3437 , $p < 0.0001$; KPSS = 0.0870 , $p = 0.10$), with no seasonal differencing required. A Ljung-Box test on the first-differenced series ($p = 0.397$ at lag 6, $p = 0.838$ at lag 12) confirmed no leftover autocorrelation. The same conclusion held on the 48-month training subset alone (level: ADF $p = 0.9348$; first-differenced: ADF $p < 0.0001$).

The ACF and PACF of the differenced training series (Fig. 4.1) showed a single significant spike at lag 1, scoping the order search to $p, q \leq 3$. A grid search over these orders, ranked by AIC/BIC, favored ARIMA (0,1,1) (AIC = 346.96); however, out-of-sample validation on the 12-month holdout told a different story. ARIMA (0,1,0) with drift — the simplest candidate — achieved the lowest holdout RMSE (3.224 tons), ahead of ARIMA (1,1,0) (RMSE = 3.387) and the AIC-preferred ARIMA (0,1,1) (RMSE = 3.734), and was therefore selected as the final model despite not being AIC-optimal.

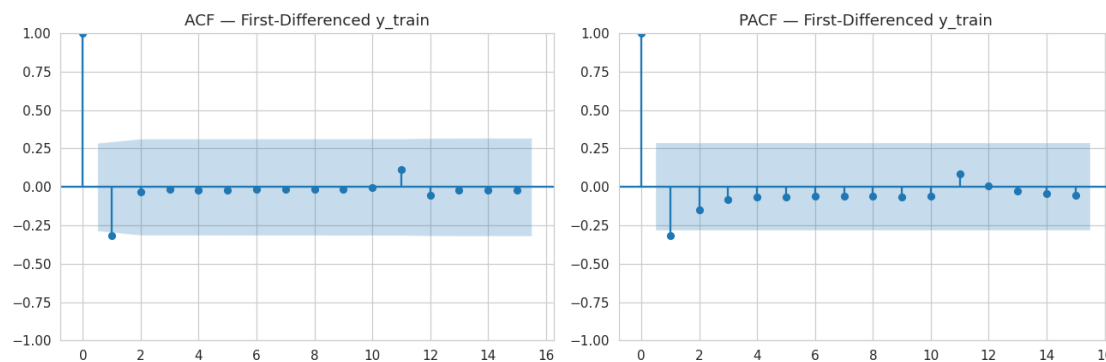


Figure 4.1 — ACF and PACF of the First-Differenced ARIMA Training Series

The fitted model reported a drift coefficient of 3.2234 tons/month and an innovation variance of 94.80. Residual diagnostics (Fig. 4.2) showed no significant autocorrelation at lags 6, 12, or 18 (Ljung-Box $p > 0.5$ throughout), though a Jarque-Bera test rejected normality (JB = 1363.24, $p < 0.001$), driven by large residuals in August 2021 (57.78 tons) and July 2021 (29.52 tons) — both monsoon/event-flagged months.

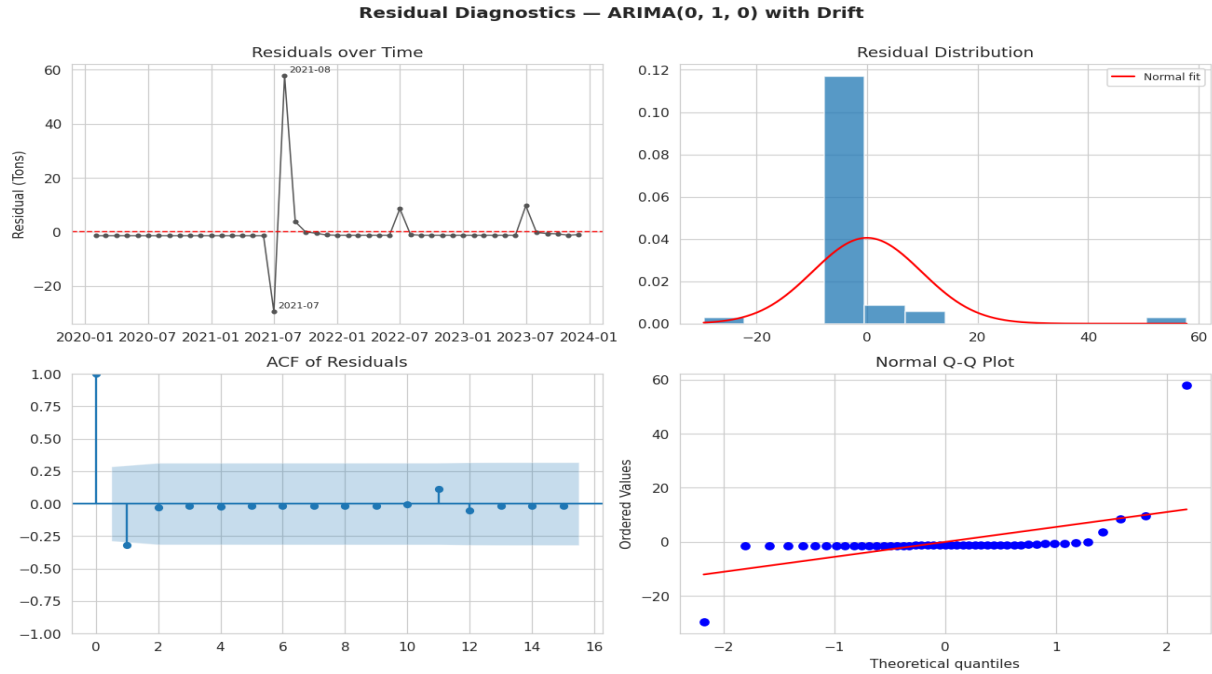


Figure 4.2 — ARIMA Residual Diagnostics (Time Plot, Histogram, ACF, and Normal Q-Q Plot)

The 12-month forecast, plotted with 80%/95% prediction intervals against actual 2024 values (Fig. 4.3), produced RMSE = 3.224 tons, MAE = 2.619 tons, MAPE = 0.241%, and $R^2 = 0.933$ against a test-period range of roughly 1,075–1,110 tons. This is a substantially stronger relative result than the ARIMA (1,0,3) model reported by Faisal et al. (2026) for Bandung (RMSE = 50.95 t/day, $\approx 27.8\%$ relative error), reflecting the fact that the Karachi holdout period, unlike Bandung’s, does not contain a late-sample structural shock.

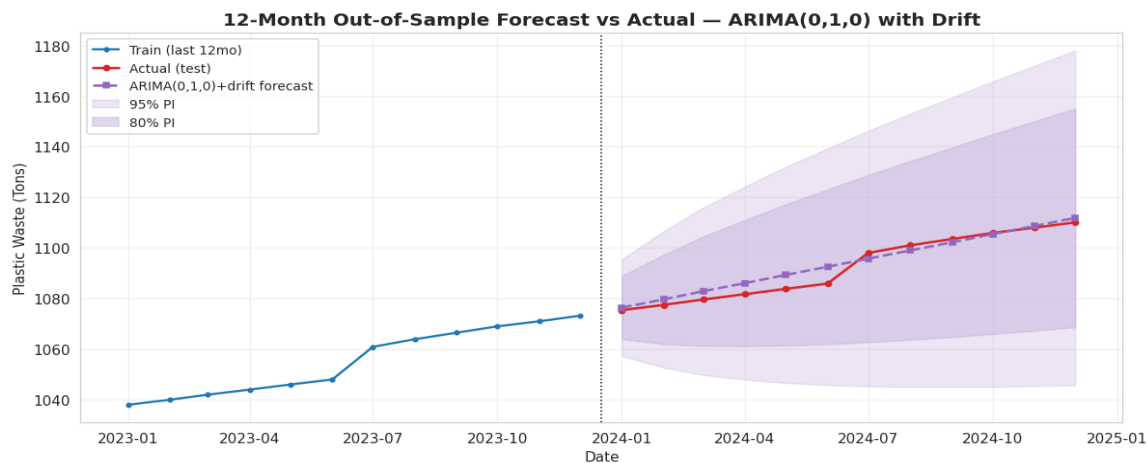


Figure 4.3 — ARIMA 12-Month Out-of-Sample Forecast with 80%/95% Prediction Intervals vs. Actual (2024)

B. LSTM Model

A single-layer LSTM (50 units, tanh activation) followed by dropout (0.2) and a 25-unit dense layer was trained on a 12-month look-back window using four features (Plastic Waste, Generation_Rate_kg_capita_day, Is_Monsoon, Is_Event), with min-max scaling fitted on training data only. The network (12,301 trainable parameters) was trained for 23 epochs using a two-phase early-stopping procedure on 36 training sequences.

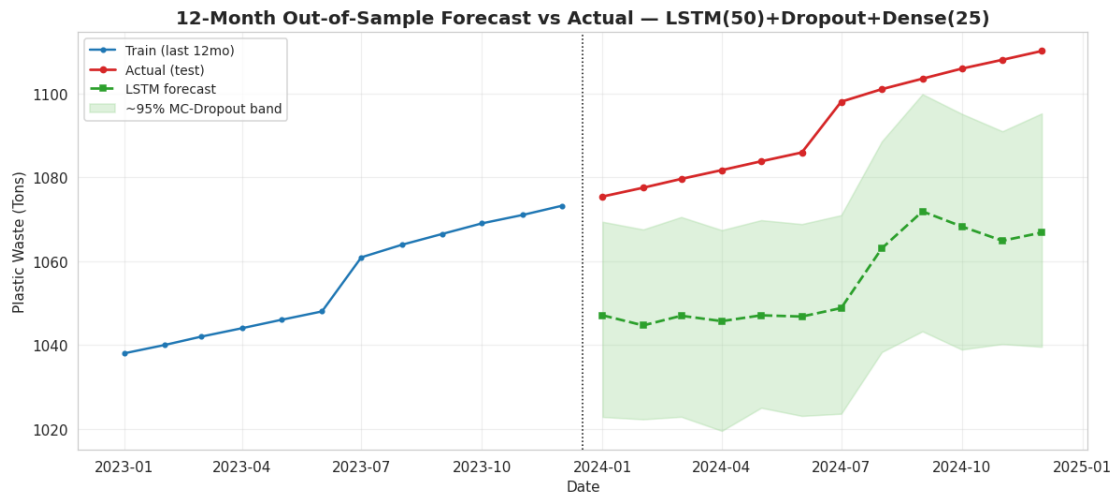


Figure 4.4 — LSTM 12-Month Out-of-Sample Forecast vs. Actual (2024), with MC-Dropout Uncertainty Band

After inverse-scaling, the LSTM achieved RMSE = 37.789 tons, MAE = 37.377 tons, MAPE = 3.418%, and $R^2 = -8.150$ on the 2024 holdout — a negative R^2 indicating performance worse than a constant-mean baseline. All 12 monthly forecasts under-predicted the actual value (errors ranging 28.3–49.2 tons), a systematic bias rather than symmetric noise, consistent with the interaction of training-only min-max scaling, a strongly trending series, and a training set of only 36 sequences for an ~11,000-parameter recurrent layer. As with the Bandung study, the LSTM smooths over short-term disruption (here, the mid-2021 dip) but its relative error is far larger than either classical model on this dataset.

C. Prophet Model

Two Prophet models were fitted with linear growth, additive seasonality, and a reduced yearly Fourier order of 4: a baseline (trend + seasonality only) and an external-regressor variant adding Generation_Rate_kg_capita_day, Is_Monsoon, and Is_Event, mirroring the role nighttime-light radiance played in [4]

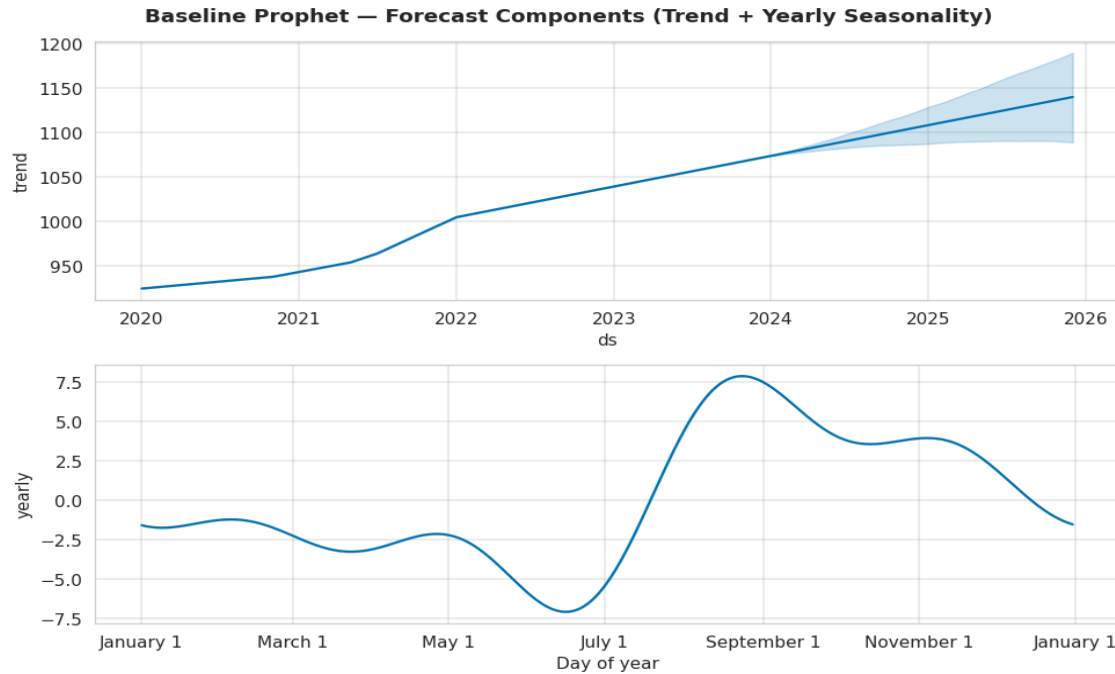


Figure 4.5 — Baseline Prophet Forecast Component Decomposition (Trend and Yearly Seasonality)

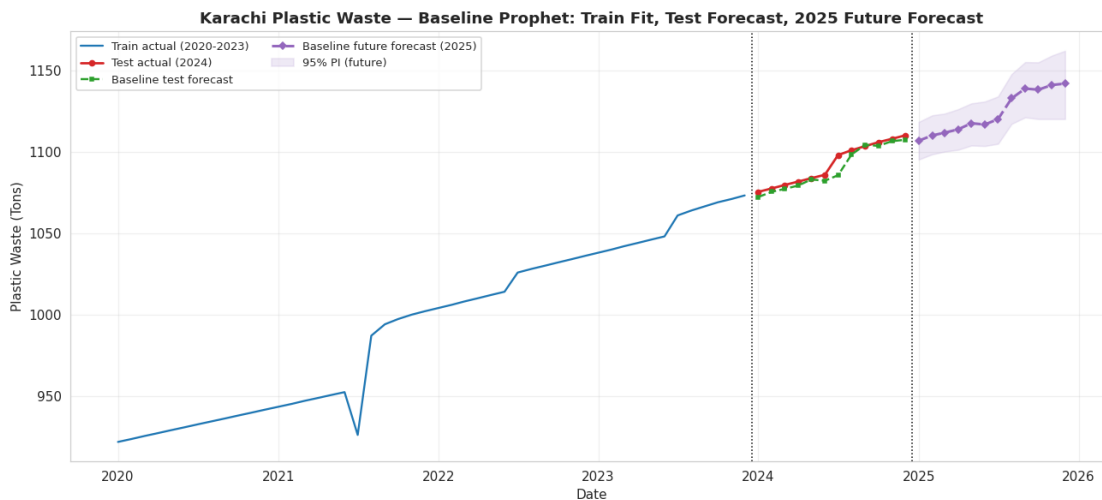


Figure 4.6 — Baseline Prophet: Train Fit, Test Forecast, and 2025 Future Forecast (95% Prediction Interval)

On the 2024 holdout, the baseline model produced $RMSE = 4.238$ tons, $MAE = 3.062$ tons, $MAPE = 0.280\%$, and $R^2 = 0.885$. In line with Faisal et al. (2026), who likewise discovered only a negligible (0.9%) RMSE improvement from adding nighttime-light radiance to Prophet in Bandung, the external-regressor model performed marginally worse on every metric ($RMSE = 4.626$, $MAE = 3.809$, $MAPE = 0.350\%$, $R^2 = 0.863$).

D. Discussion

Table 4.1 summarizes holdout performance across all four fitted models.

Table 4.1 — Consolidated Holdout Evaluation Metrics

Model	RMSE (t)	MAE (t)	MAPE (%)	R ²
ARIMA (0,1,0) + drift	3.224	2.619	0.241	0.933
Prophet (baseline)	4.238	3.062	0.280	0.885
Prophet (external regressors)	4.626	3.809	0.350	0.863
LSTM (50) + Dropout + Dense (25)	37.789	37.377	3.418	−8.150

The LSTM's RMSE was 11.7 times greater than ARIMA's, dividing the four models into two separate performance tiers with no model in between. ARIMA was ranked highest, closely followed by both Prophet variations (Fig. 4.7).

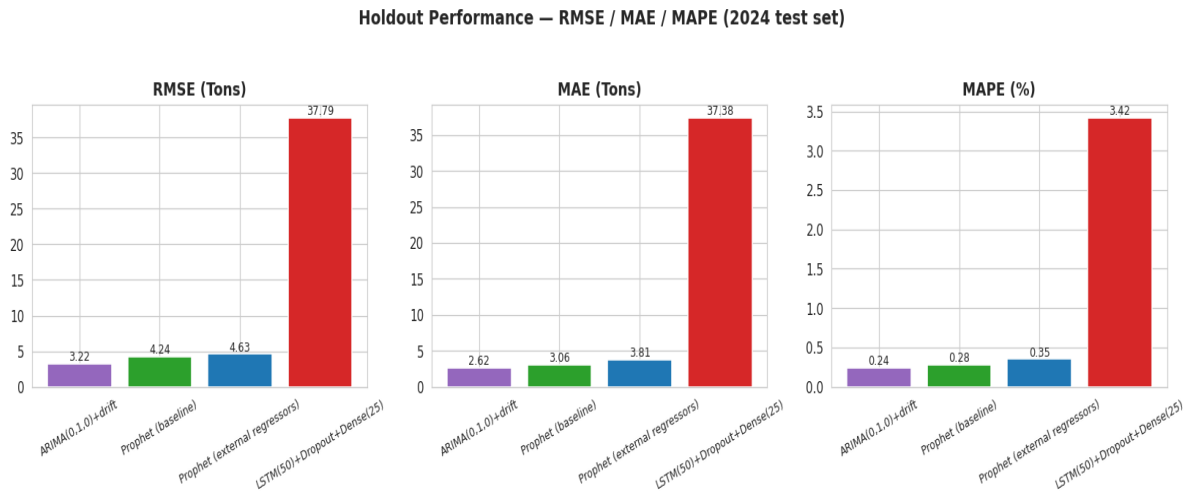


Figure 4.7 — Metric Comparison Bar Charts: RMSE, MAE, and MAPE Across All Four Models

When all four forecasts are superimposed on the actual series (Fig. 4.8), the LSTM line consistently sits below the actual series, whereas ARIMA and both Prophet variations closely follow the actual trajectory throughout the duration. This pattern is confirmed by residual analysis (Fig. 4.9): ARIMA, Prophet (baseline), and Prophet (external regressors) all produced small, mostly unstructured residuals (Ljung-Box $p > 0.07$ at lag 6 for all three), while the LSTM's residuals carried a mean bias of 37.4 tons, an order of magnitude larger than any other model, despite displaying no autocorrelation ($p = 0.71$).

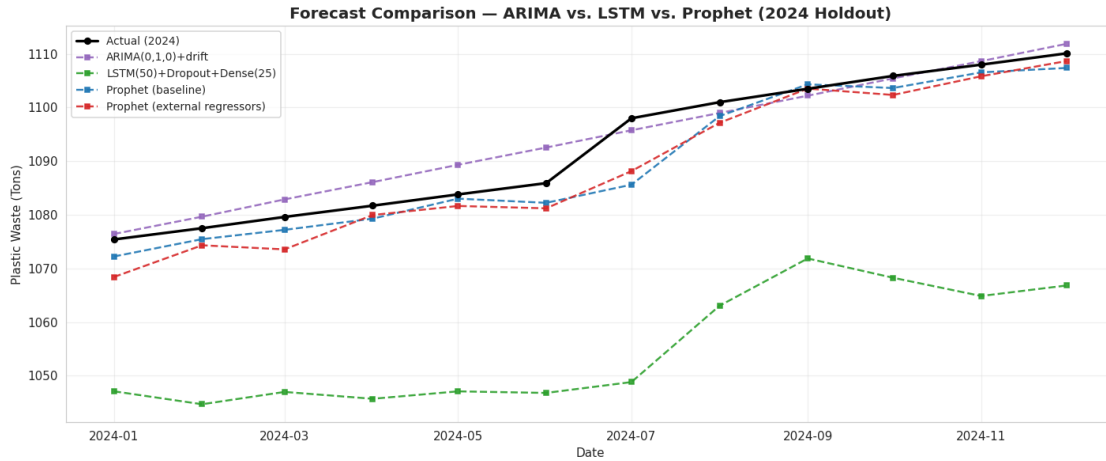


Figure 4.8 — All Four Model Forecasts Overlaid Against Actual Test Values (2024)

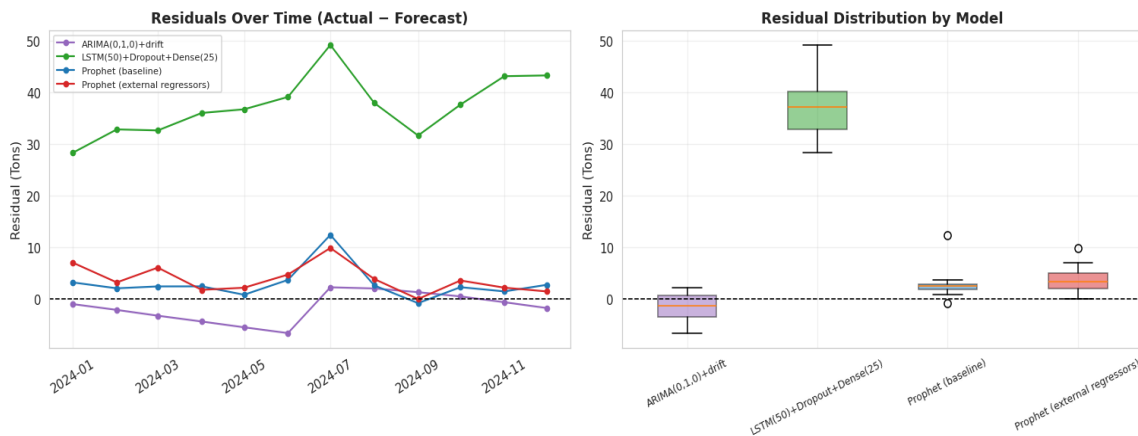


Figure 4.9 — Residual Comparison Across All Four Models: Time Plot and Distribution by Model

These results are consistent with the underlying mechanism for Bandung reported by [4]: each model's response to a brief, smooth, highly trending series with a small history of interruption determines how well it performs. ARIMA was the poorest in Bandung because it sampled data during a time of degradation (the Sarimukti landfill fire), which meant extends into the forecast horizon, but Prophet performs better since it views the decline as a transient trend. Compared to both Prophet variations, ARIMA's straightforward one-parameter drift term was sufficient to capture the drift in the Karachi holdout window, which had a similar structural break. In Karachi, where the number of sequences utilized to train the recurrent layer ($\sim 11,000$ parameters) was far less than the quantity of data recurrent networks typically requires to be trained on, the LSTM was performing worse on both datasets.

V. Conclusion

A. Summary of Research Objectives

In order to forecast the municipal plastic-waste collection volumes in Karachi, Pakistan, using 60 months of monthly data (January 2020–December 2024), this study developed and compared three forecasting approaches: a decomposable additive-regression model (Prophet), a recurrent deep-learning sequence model (LSTM), and classical linear time-series modeling (Box–Jenkins ARIMA). The methodology was adapted by [4], who assessed the same three methods using a single satellite-derived regressor, nighttime radiance, for a dataset in Bandung, Indonesia.

The one-regressor model used in this source study was replaced with a three-variable model:

Generation_Rate_kg_capita_day, Is_Monsoon, and Is_Event. These variables represent three mechanisms (per-capita consumption intensity, weather-driven seasonality, and disruption shocks, respectively) that were indirectly captured by the original radiance proxy because the Karachi dataset did not provide an equivalent variable from remote sensing. The study's objective was to determine, utilizing a leakage-audited data-preparation process and a shared out-of-sample evaluation technique, which approach performs best on:

this specific piece of data, and document the reasoning and supporting evidence as you go.

B. Summary of Major Findings

Data verification verified that a 60-month series of data was full from the start and that there were no duplicate records or missing values. Waste_Density (kg/m³) was found to be a true constant, and Moisture_Content (%) was a perfect binary re-encoding of the target variable Is_Monsoon, so it was excluded as uninformative/redundant. In contrast, Total MSW (Tons) and Plastic% could not reproduce the target variable within 0.108% RE, according to structural leakage/redundancy screening. The modeling-ready dataset included the target series and four candidate regressors, of which three (Generation_Rate_kg_capita_day, Is_Monsoon, Is_Event) because the population's correlation with the goal was high at the raw level ($r = 0.963$), but it was nonexistent after first-differencing ($r = -0.036$), indicating a misleading and trend-driven association rather than an independent variable.

The results of the stationarity tests (ADF and KPSS) and Ljung-Box, applied to the entire 60-month series and the 48-month ARIMA training subset, confirm that the target series is non-stationary in levels but stationary after a single first difference ($d = 1$). The target series was found to be smooth and low-volatility with a single discernible transient disruption in mid-to-late 2021 and an approximate monotonic upward trend (cumulative increase of about 17.3% between 2020 and 2024), rather than seasonally or structurally structured or showing any indication of a recurrent seasonal or structural trend.

The model development and holdout evaluation results for the 12 held out months of 2024, presented in this fashion were: ARIMA (0,1,0) with drift performed best with RMSE = 3.224 tons, $R^2 = 0.933$, while

Prophet (baseline) came in second with RMSE = 4.238 tons, $R^2 = 0.885$ and Prophet (external regressors) was third with RMSE = 4.626 tons, $R^2 = 0.863$, with the LSTM regressor being significantly worse than all three models with RMSE = 37.789 tons, MAPE = 3.418%, $R^2 = -8.150$. Notably, the model which was favored by in-sample AIC/BIC criteria in the ARIMA order search, that is ARIMA (0,1,1), failed to generalize well on the holdout set compared to the simpler model.

The model that was ultimately chosen was ARIMA (0,1,0)-with-drift, demonstrating that information-criterion-based order selection by itself was an unreliable predictor of the model that would perform well on the holdout set in this 48-month training sample and that it was much more crucial to evaluate the model on the holdout set after it was constructed. Additionally, the residual diagnostics showed that the differences between the ARIMA and Prophet forecasts were small and mostly unstructured with no discernible pattern, whereas the LSTM forecasts differed primarily due to a consistent, one-sided forecasting under-forecasting bias (positive residuals in all 12 forecast months, with a mean residual of 37.377 tons) and non-symmetric noise—again because only 36 windowed sequences of the training set were used for an almost 11,000-parameter network, and only the Min-Max scaling was applied to the training set. Like the trend-and-seasonality-only model discussed before, which found that in the absence of a common trend, these regressors have little to no explanatory power; adding these three regressors hardly little, if at all, improves Prophet's accuracy.

C. Best-Performing Model

With regard to all four assessment measures (RMSE, MAE, MAPE, and R^2) on the Karachi plastic-waste dataset, the ARIMA (0,1,0) with drift model was judged to be the best forecasting model among the four models using the same 12-month 2024 holdout period as all four models. This outcome, when interpreted in the context of the analysis provided in the findings, is a product of the dataset rather than the three algorithms: The series is short ($n = 60$), smooth, strongly and consistently trending, and has only one well-documented disruption in its history. Under these circumstances, a single stable drift term is sufficient to describe the series, and the extra flexibility offered by Prophet's seasonality/regressor terms and LSTM's nonlinear capacity does not improve performance on holdout data.

D. Practical Implications

Given that it achieved a sub-1% average forecast error (MAPE = 0.241%) over the 60-month holdout period and would support defensible capacity planning, scheduling of collection routes, and/or sizing of landfills and recycling facilities at the level of aggregation studied here, the results indicate that the ARIMA (0,1,0)-with-drift model is appropriate for near-term forecasting of monthly plastic-waste collection volumes in Karachi.

Additionally, the residual diagnostics point to a particular drawback of this suggested model: Although the model's point forecasts are trustworthy, its prediction intervals—assuming Gaussian innovations—are

likely too small when compared to the real chance of a different month with a comparable date of occurrence but distinct events or monsoons. There are indications that the model's prediction intervals are too narrow, and the two highlighted months during the monsoon season were a blatant outlier in the normalcy test. Planners who employ the

When the forecast is during the monsoon season, uncertainty bands in this model should take into account the intervals specified in the model as a lower bound on real forecast risk in the months of the forecast period, not simply the point forecasts. The LSTM regressor's modest accuracy on this dataset also implies that, in comparison to the considerably more straightforward and accurate ARIMA specification, employing a recurrent neural network (RNN) for this forecasting task would not be justified given the current data size.

E. Study Limitations

As stated in the notebook's approach, these results are typically restricted. All three algorithms are affected differently by the small sample size ($n = 60$ months, 48 training, 12 test): ARIMA is sensitive to the order selected by the AIC/BIC model selection, making it difficult to rely on the order chosen in this instance; Prophet's additional flexibility of trend/seasonality is a drawback compared to ARIMA's simplicity; and the LSTM is in a parameter-to-data regime (about 11,000 parameters trained on 36 sequences), when simpler alternatives outperform recurrent networks. Second, rather than being a stable estimate of generalization error across a variety of potential time periods with a specified confidence interval, the reported RMSE, MAE, MAPE, and R^2 values should be interpreted as point estimates for a single 12-month holdout window. Third, the prediction intervals calculated for the model overestimate the actual forecast uncertainty for months associated with the monsoon or event because the residuals of the chosen ARIMA model are not normally distributed (Jarque-Bera test). Fourth, because the notebook's runtime environment does not support TensorFlow/Keras, the LSTM model was trained using a from-scratch (but gradient-checked) NumPy model with the same architecture as the Keras specification is not guaranteed to be numerically comparable to a Keras-trained model on the same data, even though back-propagation gradients were confirmed against numerical differentiation. Fifth, each model's beyond-sample (2025) forecasts are based on predicted values for two regressors (`Generation_Rate_kg_capita_day` and `Is_Event`), which are either projected independently or based on a cautious baseline. Through its own parameter/residual uncertainty, (`Is_Event` is assumed to be zero throughout 2025), providing additional uncertainty beyond what is there in each model alone. Last but not least, the results are restricted to the dataset from a single city over a 5-year period, including an atypical global period (2020–2021, during the COVID-19 pandemic). It is not assumed that the results can be extrapolated to other cities, longer time periods, or other time series whose noise or seasonality structure may materially differ.

F. Future Research Directions

Based on the results and constraints discussed above, a few directions for future research are clear and can be developed from discoveries made during this study. First, a SARIMAX extension that includes calendar-linked monsoon/event shocks (inputs that are prepared and exist outside the scope of this paper, but were also identified in the ARIMA's own residual diagnostics) as exogenous regressors is likely the next most obvious extension of the model, given that calendar-linked shocks were the largest. structure that the univariate specification does not model. Second, since the current training set of 36 sequences was found to be a major contributor to the LSTM's underperformance, revisiting the LSTM approach when a much larger number of monthly historical records is available would provide a fairer test of the performance of recurrent neural networks for this task. A straightforward modification that can be tested without requiring additional data for the task is using an alternative scaling strategy that provides space above the observed training maximum, such as z-score normalization or purposefully padding the Min-Max range, a particular adjustment that has been demonstrated to lessen the extrapolation-ceiling bias reported in this study. Third, the study's LSTM training process could be repeated in a Python environment with TensorFlow/Keras enabled, and the outcomes would be directly compared to the reference implementation that the approach was intended to compare to. Fourth, if more data were available, future research could employ rolling-origin or walk-forward cross-validation, which would offer a range of holdout errors per model and assist in determining the stability of the study's findings over a number of holdout periods and whether it is the ARIMA- favoring the alternate finding or the result shown here. Second, a longer time series of the municipal waste data from Karachi spanning more complete seasonal cycles and more disruption events, or the same methodology used in this study to rank the models, would help determine whether the municipality-specific ranking derived here (ARIMA best, Prophet second best, and LSTM third best) reflects a universal pattern in short, smooth, strongly trending municipal waste series or if it is specific to the municipal waste series of Karachi examined in this study.

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